Linux: Securing File Permissions

Enforcing least privilege on a research team's project directory

Type	Hands-on lab (Linux command line)
Tools	Bash, `ls`, `chmod`
Skills	Linux file system, permission strings, symbolic and octal `chmod`, least privilege
Source	Google Cybersecurity Professional Certificate, Course 4 (*Tools of the Trade: Linux and SQL*)

Project description

The research team's files in the `~/projects` directory had permissions that didn't match the access each user should have. I audited the current permissions and used `chmod` to apply least privilege: nobody gets more access than their role needs.

1. Check file and directory details

```
researcher2@5d738f0f927b:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Dec  2 15:27 .
drwxr-xr-x 3 researcher2 research_team 4096 Dec  2 15:27 ..
-rw--w---- 1 researcher2 research_team   46 Dec  2 15:27 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Dec  2 15:27 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Dec  2 15:27 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Dec  2 15:27 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Dec  2 15:27 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Dec  2 15:27 project_t.txt
researcher2@5d738f0f927b:~/projects$
```

Listing all files, including hidden ones, with permissions

`ls -la` lists every item in the directory in long format (`-l`), including hidden files (`-a`). The output shows **one directory (`drafts`)**, **one hidden file (`.project_x.txt`)** and **four project files** (`project_k`, `project_m`, `project_r` and `project_t`). The first column is a 10-character permission string for each item.

2. Reading the permission string

Position	Meaning
1	File type: `d` = directory, `-` = regular file
2–4	**User** (owner) permissions: read `r`, write `w`, execute `x`
5–7	**Group** permissions
8–10	**Other** (everyone else) permissions

A `-` in positions 2–10 means that permission isn't granted. **Example:** `project_t.txt` is `-rw-rw-r--`. It's a regular file; the user and group can read and write it; others can only read it; and no one can execute it.

3. Remove write access for "other"

Policy: "other" must not have write access to any file. The audit showed `project_k.txt` was `-rw-rw-rw-`, which means anyone on the system could change it.

```
researcher2@5d738f0f927b:~/projects$ chmod o-w project_k.txt
researcher2@5d738f0f927b:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Dec  2 15:27 .
drwxr-xr-x 3 researcher2 research_team 4096 Dec  2 15:27 ..
-rw--w---- 1 researcher2 research_team   46 Dec  2 15:27 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Dec  2 15:27 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Dec  2 15:27 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Dec  2 15:27 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Dec  2 15:27 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Dec  2 15:27 project_t.txt
researcher2@5d738f0f927b:~/projects$
```

Removing write permission for other on project_k.txt

`chmod o-w project_k.txt` removes (`-`) write (`w`) permission from other (`o`). Running `ls -la` again confirms the file is now `-rw-rw-r--` (octal 664).

4. Lock down a hidden archived file

Policy: `.project_x.txt` has been archived. No one should be able to write to it, but the user and group should be able to read it.

```
researcher2@3213bbc1d047:~/projects$ chmod u-w,g-w,g+r .project_x.txt
researcher2@3213bbc1d047:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Dec 20 15:36 .
drwxr-xr-x 3 researcher2 research_team 4096 Dec 20 15:36 ..
-r--r----- 1 researcher2 research_team   46 Dec 20 15:36 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Dec 20 15:36 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Dec 20 15:36 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Dec 20 15:36 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Dec 20 15:36 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Dec 20 15:36 project_t.txt
researcher2@3213bbc1d047:~/projects$
```

Updating permissions on the hidden file .project_x.txt

`chmod u-w,g-w,g+r .project_x.txt` removes write from the user and group and adds read for the group. The result is `-r--r-----` (octal `440`; the equivalent command is `chmod 440 .project_x.txt`). The leading `.` in the name is what makes it hidden. *(This screenshot was taken in a separate lab session, so `project_k.txt` still shows its original permissions.)*

5. Restrict a directory to its owner

Policy: only `researcher2` should be able to access `drafts` and its contents. The directory was `drwx--x---`, which meant the group could enter it.

```
researcher2@5d738f0f927b:~/projects$ chmod g-x drafts
researcher2@5d738f0f927b:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Dec  2 15:27 .
drwxr-xr-x 3 researcher2 research_team 4096 Dec  2 15:27 ..
-r--r----- 1 researcher2 research_team   46 Dec  2 15:27 .project_x.txt
drwx------ 2 researcher2 research_team 4096 Dec  2 15:27 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Dec  2 15:27 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Dec  2 15:27 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Dec  2 15:27 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Dec  2 15:27 project_t.txt
researcher2@5d738f0f927b:~/projects$
```

Removing group execute permission from the drafts directory

On a directory, execute (x) is the permission to enter it. `chmod g-x drafts` removed this from the group. `researcher2` already had full access, so the directory is now `drwx------` (octal `700`).

Summary

Item	Before	After	Command
`project_k.txt`	`-rw-rw-rw-`	`-rw-rw-r--`	`chmod o-w project_k.txt`
`.project_x.txt`	`-rw--w----`	`-r--r-----`	`chmod u-w,g-w,g+r .project_x.txt`
`drafts/`	`drwx--x---`	`drwx------`	`chmod g-x drafts`

I audited the directory with `ls -la` and corrected three items so the permissions match the organisation's authorisation policy. Overly broad permissions, such as files anyone can write to, are a common way for attackers to tamper with data or escalate privileges, so checks like this are routine in system hardening.